# SANGEON JEON

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## Education

**Georgia Institute of Technology** **Atlanta, GA**
*Ph.D. in Electrical and Computer Engineering* (Advisor: Xiaoli Ma) *Jan. 2026 – Present*
*M.S. in Electrical and Computer Engineering* [GPA 4.0 /4.0] *Jan. 2025 – Dec. 2025*
*B.S. in Electrical and Computer Engineering* [GPA 4.0 /4.0] *Jan. 2023 – Dec. 2024*

**Korea Advanced Institute of Science and Technology (KAIST)** **Daejeon, South Korea**
*B.S. in Electrical Engineering* [GPA 3.95 /4.3] *Feb. 2018 – Feb. 2025*

## Experience

**6G Physical Layer (PHY) OCDM Synchronization & Parameter Estimation**, Xiaoli Ma Lab, Georgia Tech *Sept. 2025 – Present*

- Developed a robust preamble for Orthogonal Chirp Division Multiplexing (OCDM), a leading 6G candidate
- Formally proved the identifiability and decoupling of joint Carrier Frequency Offset (CFO), Timing Offset (TO), and channel parameters, ensuring no ambiguity in parameter estimation.

**Joint TIADC Calibration and Channel Estimation for Wideband Systems**, SIC-X Lab, KAIST *Aug. 2024 - Dec. 2025*

- Devised a Chebyshev polynomial approximation-based Maximum Likelihood (ML) estimator for the joint estimation of channel taps and mismatches in Time-Interleaved Analog-to-Digital Converters (TIADCs) under OFDM transmission.
- Replaced exhaustive iterative searches with an efficient polynomial rooting approach, achieving a 4x speedup in execution while significantly mitigating mismatch-induced distortion.

**Advanced Bayesian Inference for Global Disease Burden**, Collaboration with Kyung Hee Univ. & WFH *Jan. 2025 - Dec. 2025*

- Pioneered a framework to infer joint sex-by-age distributions from marginal reports (e.g., reconciling independent sex-specific and age-specific data) and handled unlabeled data (unknown sex/age).
- Developed an initial EM-based splitter and subsequently superseded it with a unified Bayesian engine to eliminate error propagation between stages, achieving more robust convergence.
- Drastically narrowed Uncertainty Intervals (UI) by integrating the splitting process directly into a PyMC5-based probabilistic model, maximizing estimation precision from incomplete datasets.
- Modeled global prevalence for rare diseases across 204 countries; validated reliability through extensive synthetic data simulations and cross-validation.

**KAIST Maker-a-thon (1st Prize / $2,000 Award) [link]** *Jul. 2024*

- Engineered a user-centric RFID-based food management system; reduced hardware cost and logic complexity by relocating sensors to token containers instead of the refrigerator.
- Led a 4-member team through rapid prototyping; managed the entire software architecture and coding using C# UWP.

**Bio-signal Recording & Synthesis System**, Collaboration with Vivent, Switzerland **[link]** *Jan. - Apr. 2024*

- Built an M2M system to record and synthesize Venus flytrap action potentials; developed a time-delay correction algorithm for reliable 100 Hz signal reconstruction, triggering plant response electrically.

**Software Engineer Intern | *Dong Il Vision* | Anyang, South Korea** *Jun. – Aug. 2023*

- Developed a C# WPF-based labeling program for machine learning, optimizing the computer vision data pipeline.

## Publication

Y. D. Jeong*, Y. Son*, **S. Jeon* (Co-first)**, et al., "Global prevalence of obsessive-compulsive and related disorders: a systematic review and modelling study," ***The American Journal of Psychiatry***, (Minor Revision), 2026.

Y. D. Jeong*, Y. Son*, **S. Jeon* (Co-first)**, et al., "Global prevalence of spinal muscular atrophy (1960-2024): a Bayesian modelling study," ***Pediatric Neurology***, (Minor Revision), 2026.

## Skills

**Programming:** MATLAB(Expert), Python(PyMC5, Pandas, Pytorch, NumPy, SciPy), C#(WPF/UWP), C++, Java
**Tools:** HFSS, Solidworks, NI Multisim, 3D Printing, Laser Cutting, CNC, Lab Equipment
**Languages:** Korean (Native), English (Full Professional Proficiency), Japanese (Conversational)